

SMART REAL-TIME AGE AND EMOTION-BASED ADVERTISING IN MALLS USING CUSTOM-BUILT MODELS

Project ID: 25-26J-091

Project Proposal Report

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Sri Lanka

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REAL-TIME POSE-BASED ENGAGEMENT DETECTION AND BEHAVIORAL ANOMALY INTEGRATION SYSTEM

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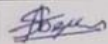
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Declaration

Declaration

I declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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2025/08/25
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Abstract

This research aim to develop a smart, advertising system for shopping malls that dynamically chooses and displays relevant advertisements in real time based on the age group, emotional state, and behavior of the audience, while maintaining strict privacy regulations. Current digital signage solution in Sri Lanka and similar markets often displays static or scheduled advertisements, failing to adapt content to the actual audience present. This results in lower engagement and missed marketing opportunities. The study addresses the research problem of integrating demographic detection, emotion recognition, and behavioral analytics into a unified, privacy-first advertising system.

The proposed system architecture incorporates pose-based engagement detection, behavioral anomaly analysis, and a dynamic ad recommendation engine. Data is gathered from camera infrastructure and processed using real-time computer vision algorithms without storing identifiable facial information. Engagement level are calculated using body orientation and dwell time, while anomalies such as loitering or erratic movements are identified for both marketing and the security purposes.

The project will be implemented using machine learning models, computer vision methods, and cloud-based processing for scalability. Key pillars include privacy-preserving analytics, multi-purpose data utilization, and real-time audience targeting. The methodology involves requirement analysis, model training, system integration, and performance testing in simulated retail environments before the deployment.

The anticipated outcome is an integrated retail analytics solution capable of boosting advertisement engagements, enhancing customer experience, and providing actionable security insights—all without compromising privacy. This research contribute to advancements in retail systems, context-aware marketing, and ethical surveillance technologies, offering a scalable framework for both local and global retail applications.

Keywords: real-time engagement detection, behavioral analytics, privacy-preserving computer vision, pose based detection

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Introduction

Artificial intelligence (AI) and computer vision are transforming retail analytics, enabling businesses to better understand customer behavior, deliver targeted advertisements, and improve security operations. Facial expression recognition systems have been built to adjust advertisements in real time based on detected emotional states [1], [2]. Crowd counting and density estimation models, such as multi-column convolutional neural networks (MC-CNN), have been applied to measure audience size and adapt marketing strategies accordingly [3]. Pose estimation frameworks have been explored as privacy-preserving tools to infer audience attention levels without capturing identifiable facial data [4]. In the security domain, behavioral anomaly detection techniques—such as trajectory analysis and temporal pattern modeling—have been used to identify unusual or suspicious activities in retail environments [5], [6].

In Sri Lankan shopping malls, however, digital advertisements remain largely static or follow fixed schedules, showing the same content regardless of who is present. This mismatch between audience characteristics and displayed content reduces viewer engagement and diminishes the overall impact of advertising. Modern consumers, accustomed to personalized and interactive digital experiences, increasingly expect advertisements to be relevant, timely, and emotionally engaging. These studies show that individual methods for demographic detection, emotion analysis, engagement measurement, and anomaly detection can be effective. However, most existing works implement them as separate systems targeting either marketing or security. Many also rely on biometric identifiers, such as facial recognition, which raise privacy and ethical concerns, especially in public retail spaces. State-of-the-art systems are now capable of performing real-time, multi-person tracking, emotion detection, and dynamic advertisement delivery in crowded environments.

The Real-Time Pose-Based Engagement Detection and Behavioral Anomaly Detection component measures audience attention using body orientation and dwell time, detects unusual behaviors, and then feeds the results into the group's adaptive advertisement recommendation engine, allowing dynamically change the content based on real audience behavior. By integrating marketing and security analytics in a privacy-first system, this research advances the state of the art, delivering a scalable, ethically responsible retail analytics solution for Sri Lankan shopping malls and similar public spaces worldwide.

Background & Literature Review

In the modern retail environment, digital advertising plays a critical role in shaping customer experiences and influencing purchasing decisions. The increasing presence of large-format digital display in shopping malls provide retailers with opportunities to deliver brand messages to diverse audiences. However, in many regions, including Sri Lanka, these advertisements are still static or follow fixed schedules, with little consideration for who is present at any given moment. This results in a lack of contextual relevance, reduced engagement, and missed chances for personalized customer interaction.

Globally, there has been a shift towards smart, dynamic advertising system capable of dynamically selecting and displaying contents based on real-time audience analysis [7]. These systems leverage advances in computer vision, machine learning, and behavioral analytics to analyze demographic attributes, emotional states, and audience behaviors from camera feeds, and adapt advertisements accordingly. Such technology aligns with the growing demand for personalized, interactive experiences that mirror the targeted advertising common in online platforms [8].

The proposed research aim to develop a real-time, privacy-preserving, dynamic advertising system for shopping malls. By analyzing age group, emotional state, and behavioral patterns from camera feeds, the system will dynamically select and display the most relevant advertisement from a pre-categorized database. This ensures higher engagement, better alignment between content and audience mood, and compliance with privacy requirements by avoiding the storage of identifiable personal data and also public safety.

2.1 Audience Demographic Analysis

Estimating the age group of individuals in a crowd is a foundational element in targeted advertising. Early approaches relied on handcrafted facial features, while recent research has shifted towards deep convolutional neural networks (CNNs), which can automatically extract robust features from images and operate in real time [9]. Although highly accurate, the use of identifiable facial data has raised ethical and legal concerns [10]. Privacy-first approaches have

emerged, where demographic estimation is performed in real time without retaining identifiable images, making them more suitable for public retail settings.

2.2 Emotion Recognition in Marketing

Emotion recognition is used to determine the audience's emotional state—such as happiness, sadness, or neutrality based on facial expressions. Studies show that tailoring advertisements to match the audience's mood can significantly improve brand recall and engagement [11]. CNN-based emotion recognition models have demonstrated high accuracy in both controlled and real-world conditions. However, in public spaces, privacy concerns and variable lighting conditions must be addressed through ethical, privacy-compliant processing methods.

2.3 Behavioral and Engagement Analysis

Behavioral analytics provide insights into customer interests and attention through the study of movement, dwell time, and body orientation. Pose estimation techniques, such as Open Pose [4], can detect skeletal keypoints to infer whether a person is engaged with a display, without using facial identifiers. Research indicates that body orientation combined with dwell time is a strong predictor of visual attention [12]. Such methods also allow for behavioral anomaly detection, identifying patterns such as loitering or erratic movement, which can provide valuable security insights alongside marketing data [6].

2.4 Dynamic Content Delivery Systems

Dynamic content delivery in advertising involves selecting and displaying advertisements based on live audience analysis. Early systems relied on pre-programmed schedules, while AI-driven systems now use real-time analytics to adapt to the current audience profile [13]. Studies have shown that dynamic systems yield significantly higher engagement rates and sales conversions compared to static displays [14]. This is achieved by matching advertisements categories (e.g., youth-oriented, family-focused) with the audience's demographic and emotional characteristics.

2.5 Privacy and Ethical Considerations

The use of biometric and behavioral data in public advertising introduces privacy challenges, especially in jurisdictions with strict data protection laws such as GDPR [15]. Public backlash against facial recognition in advertising campaigns has further emphasized the need for privacy-by-design solutions [16]. Techniques such as anonymized skeletal tracking, on-device processing, and immediate feature extraction without storage have been recognized as best practices for ensuring compliance and maintaining consumer trust.

This system will allow Sri Lankan shopping malls to transition from static advertising to responsive, context-aware marketing, improving customer experiences and increasing the return on investment for advertisers.

Research Gap

Feature	Existing Systems (Research)	Proposed Application Component
Privacy-Preserving Engagement Detection	Many engagement detection systems rely on facial recognition or gaze tracking, raising privacy concerns and limiting adoption in public retail spaces [1], [2].	Use anonymized pose estimation (body orientation + dwell time) to calculate an Engagement Index without storing biometric data, ensuring full privacy compliance.
Unified Engagement and Anomaly Detection	Engagement detection and anomaly detection are typically implemented as separate systems, requiring independent hardware and processing pipelines [17], [3], [6].	Develop a single integrated module that simultaneously tracks engagements and detects behavioral anomalies, minimizing infrastructure costs and enabling cross-functional analytics.
Real-Time Multi-Person Tracking in Crowded Environments	Some systems can detect engagement or anomalies, but performance drops significantly in high-density crowds or with occlusions [4], [5].	Implement robust, real-time multi-person tracking optimized for crowded shopping mall environments using pose estimation and temporal movement analysis.
Integration with Dynamic Ad Recommendation	Previous works often stop at analytics, without directly connecting into content delivery systems [13].	Feed engagement and anomaly metrics directly into the group's adaptive ads recommendation engine, so displayed content updates dynamically in real time.

Dual-Purpose Marketing & Security Insights	Security analytics and marketing analytics are usually siloed, missing opportunities for shared intelligence [14].	Provide dual-purpose insights—helping marketers optimize ad content while enabling security teams to respond to unusual behavior without extra infrastructure.
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Table 1 : Research gap

The relationship between audience engagement analysis, behavioral anomaly detection, and adaptive advertising reveals a notable gap in creating a single, integrated system that can serve both marketing and security objectives in real time. Existing solutions predominantly focus on either engagement measurement or anomaly detection, with little attempt to merge the two into a unified, privacy-preserving architecture. Furthermore, many existing systems depends on facial recognition, creating privacy concerns that limit their deployment in public retail spaces.

This research aims to fill these gaps by developing an integrated, real-time module that calculates an Engagement Index using anonymized pose data, detects behavioral anomalies, and feeds both metrics directly into a dynamic ad recommendation engine. The innovative aspect lies in combining these traditionally separate analytics streams into a dual-purpose tool capable of enhancing both customer engagement and security monitoring.

By avoiding biometric storage, operating effectively in crowded environments, and enabling instant advertisement adaptation based on live audience engagements, this component introduces a novel approach that builds on existing pose estimation and anomaly detection research while addressing practical, ethical, and operational challenges in modern retail analytics.

Research Problem

In the modern retail landscape, shopping malls have increasingly adopted digital displays as a primary medium for advertising. However, in Sri Lanka, most of these displays remain static or operate on fixed schedules, showing the same content regardless of the demographic profile, mood, or engagement of the audience present at any given moment. This results in a lack of contextual relevance, reduced audience engagement, and missed opportunities for targeted marketing.

Global studies have shown that dynamic, personalized advertising common in online platforms can significantly improve consumer attention, brand recall, and purchase intent when adapted to the audience's characteristics in real time. However, the adoption of such intelligent, adaptive advertising systems in physical retail spaces within Sri Lanka remains minimal.

Although individual techniques like pose estimation for engagement measurement [4], dwell-time analysis, and behavioral anomaly detection [6] have been studied, they are often applied as standalone systems. These systems either focus only on marketing analytics (e.g., measuring attention levels) or security monitoring but rarely combine the two. Moreover, many existing systems depend on privacy-invasive methods such as facial recognition or gaze tracking, which raise ethical concerns in public spaces.

There is a clear research gap in the development of a real-time, privacy-preserving, advertising system that combines demographic analysis, emotional state detection, engagement measurement, and behavioral anomaly detection to dynamically adapt advertisements while simultaneously offering security insights.

The challenge lies in designing a system capable of:

- Integrating multiple analytics streams (age group, mood, body orientation, dwell time, anomaly detection) into a single, real-time framework.
- Ensuring privacy compliance through anonymized data processing and avoidance of identifiable biometric storage.

- Accurately measure engagements through body orientation and dwell time, without relying on identifiable facial features.
- Detect behavioral anomalies such as loitering, crowd surges, or erratic movement to support both marketing and security functions.
- Operating effectively in crowded mall environments, where lighting variation, and high-density movement can degrade system performance.
- Feeding analytics directly into a dynamic ad recommendation engine to enable immediate content adaptation that maximizes relevance and engagements.

By addressing these challenges, the proposed research aims to create a scalable, ethically responsible retail analytics platform that bridges the gap between marketing effectiveness and public safety, setting a new standard for intelligent, audience-aware advertising in Sri Lankan shopping malls and beyond.

Objectives

Main Objective

To develop a smart, real-time advertising system for shopping mall that dynamically selects and display relevant advertisements based on real-time analysis of audience characteristics—specifically age group, emotional state, and behavioral engagement—while ensuring privacy, relevance, and improved viewer engagements.

Specific Objectives

- **Implementation of Real-Time Multi-Person Pose Estimation**

Develop and deploys advanced pose estimation techniques capable of accurately extracting anonymized skeletal key point (head, shoulders, torso) from camera footages in real-time, facilitating non-intrusive behavior monitoring.

- **Design and Validation of a Continuous Engagement Index**

Create a customized engagement scoring metric that fuse body orientation, dwell time near display, and stillness to generate continuous, per-person engagement scores corresponding to individual advertisement screens.

- **Integration of Behavioral Anomaly Detection**

Incorporate pose-based anomaly detection method to identify and flag suspicious shopper behaviors such as potential shoplifting or tampering, enabling timely security intervention without compromising individual privacy.

- **Ensuring Privacy Compliance and Ethical Data Handling**

Restrict all data processing to anonymized pose key points only, explicitly excluding facial or other biometric data, to comply fully with GDPR and ethical AI principles, thereby safeguarding shopper privacy.

- **Real-Time Data Communication with Advertising Engine**

Develop seamless interfaces to transmit continuous engagement score to an adaptive advertising engine that dynamically select or updates advertisement content based on live shopper attention signals.

Methodology

System Architecture Diagram

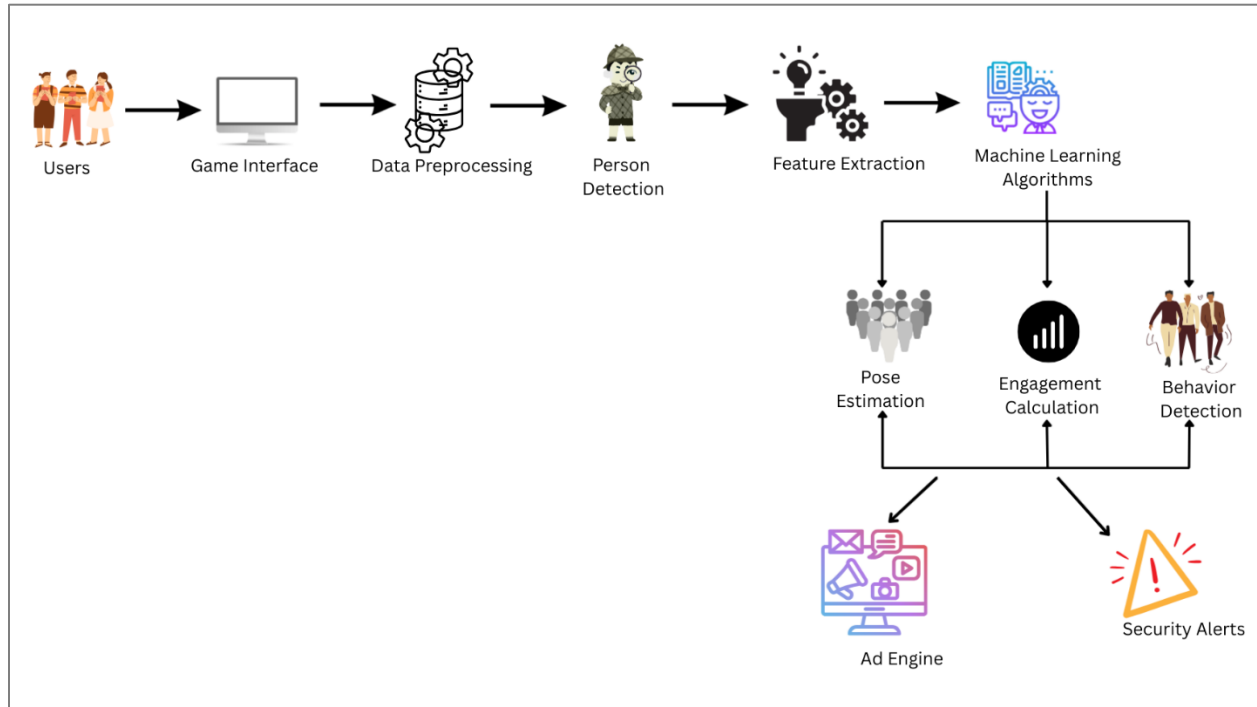


Figure 1: System Architecture Diagram of the Component

The proposed system for real-time engagement detection and behavioral anomaly integration in shopping malls focuses on utilizing camera infrastructure without the need for facial recognition or other invasive biometric tools. The approach begins by capturing live video feeds from screen cameras and processing them through computer vision modules to extract pose estimation, body orientation, and dwell time data. This allows the system to compute an Engagement Index, representing the audience's attention toward advertisements. The engagement data are then fed into the dynamic advertisement recommendation engine, which selects the most relevant content from a pre-categorized database and displays it instantly on mall screens.

Simultaneously, behavioral anomaly detection algorithms analyze movement patterns, identifying unusual activities such as loitering, erratic movement, or crowd surges. These detections are processed in real-time and logged into the system for both marketing insights and security alerts. Security-relevant anomalies are sent to the monitoring dashboard for immediate review by mall security personnel.

This methodology ensures privacy-first analytics by only using anonymized skeletal tracking data, without storing identifiable images or facial features. The system runs on cloud-based infrastructure for scalability and real-time responsiveness.

Understanding the Application of Key Pillars

In this project, the key pillars include computer vision, behavioral analytics, and real-time adaptive content delivery. Each plays a crucial role in achieving the main objective of measuring audience engagement, detecting behavioral anomalies, and dynamically updating advertisements in a privacy-preserving manner.

1. **Computer Vision:** Utilized for real-time pose estimation to detect skeletal keypoints, determine body orientation, and calculate dwell time without capturing or storing identifiable facial data. This ensures privacy compliance while still enabling accurate measurement of audience attention. The output of this process forms the basis of the Engagement Index a quantitative measure of how engaged viewers are with displayed advertisements.
2. **Behavioral Analytics:** Focuses on analyzing movement patterns to detect unusual or suspicious behaviors, such as loitering, erratic movement, or crowd surges. These insights serve a dual purpose: they provide marketers with valuable data on low-engagement periods and alert security teams to potential risks, enabling a multi-functional, cost-efficient system.
3. **Cloud Computing:** Employs scalable cloud infrastructure for real-time processing, data storage, and seamless integration with digital signage systems. This ensures the system can scale across multiple retail locations while maintaining responsive adaptive advertising and security monitoring.
4. **Real-Time Adaptive Content Delivery:** Integrates engagement and anomaly metrics into a dynamic advertisement recommendation engine capable of instantly updating content based on real audience behavior. This maximizes advertisement relevance and audience engagement while creating an interactive, responsive retail marketing environment.

Mathematical Equations

1. Centroid (per person)

Use the torso/hip keypoints to get a stable person location

$$c_t = \left(\frac{x_{hipL,t} + x_{hipR,t}}{2}, \frac{y_{hipL,t} + y_{hipR,t}}{2} \right)$$

2. Orientation (body-facing vector) and orientation angle

Body-facing vector (shoulders $\rightarrow$ neck or shoulders midpoint $\rightarrow$ neck)

$$u_t = p_{neck,t} - \frac{p_{shoulderL,t} + p_{shoulderR,t}}{2}$$

Vector to screen center s (in image coords) and angle

$$\cos \theta_t = \frac{u_t \cdot s}{|u_t| |s|}, \quad \theta_t = \arccos(\cos \theta_t)$$

3. OrientationScore

Clip negative facing-away values

$$OrientationScore_t = \max(0, \cos \theta_t)$$

4. Dwell time & DwellScore

$$T = t_{out} - t_{in}$$

$$DwellScore = \min\left(1, \frac{T}{T_{ref}}\right)$$

5. Engagement Index (E-Index)

Combine orientation + dwell (weights w_1, w_2 , $w_1 + w_2 = 1$)

$$E = w_1 \cdot OrientationScore + w_2 \cdot DwellScore$$



Figure 2 : Skeletal tracking to measure engagement.

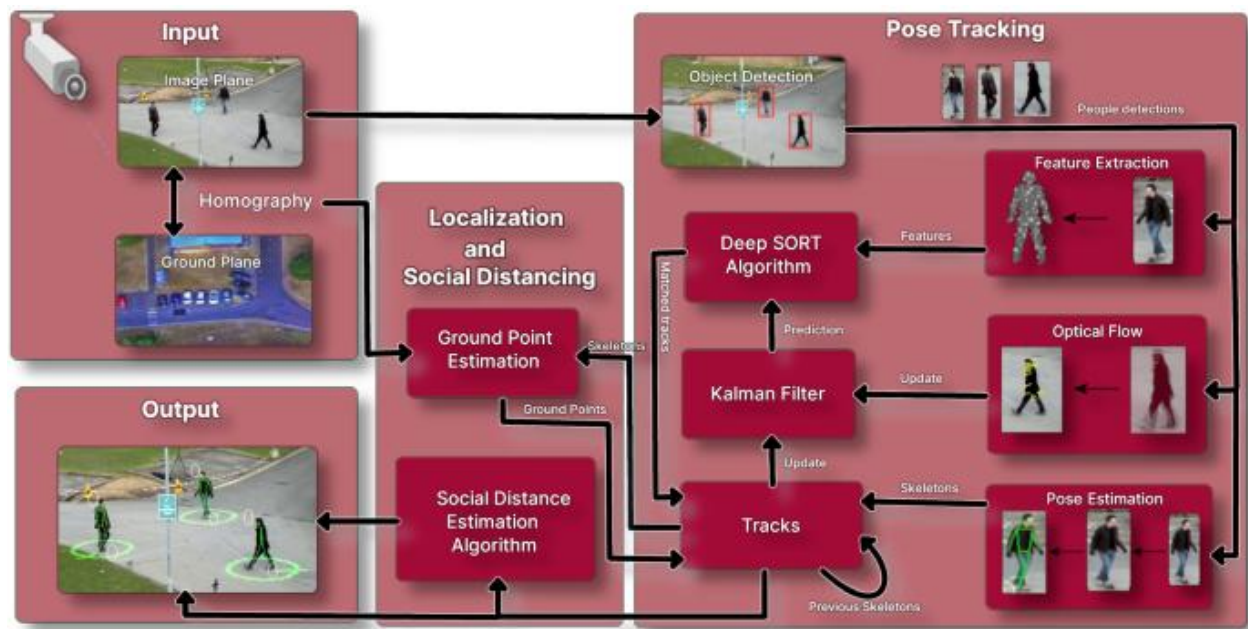


Figure 3 : Pose estimation model identifying audience orientation and attention levels.

Application of Technologies in the Relevant Area

Computer Vision Technologies:

- **Algorithms:** Real-time pose estimation algorithms will be used to detect skeletal keypoints, body orientation, and dwell time. Detects the body orientation and engagement levels of shoppers from camera video feeds by identifying key body joints like head, shoulders, and torso. This helps understand where and how shoppers pay attention without using facial recognition. Object detection and tracking methods will support multi-person tracking in crowded mall environments.
- **Tools and Libraries:** Frameworks such as OpenCV, TensorFlow, PyTorch, and MediaPipe will be employed for pose estimation, skeleton tracking, and anomaly detection tasks.

Behavioral Analytics Technologies:

- **Algorithms:** Spatiotemporal analysis and unsupervised anomaly detection methods (e.g., Isolation Forest, Autoencoders, LSTM networks) will be used to identifies unusual behaviors such as loitering, erratic movements, or suspicious activities by analyzing movement patterns over time, enhancing retail security.
- **Tools and Libraries:** Libraries like Scikit-learn, Keras, Pandas, and NumPy will be used for data preprocessing, anomaly detection, and behavioral pattern recognition.

Security Alert System

- Sends instant notifications to security teams about detected anomalies for quick response and preventive actions.

Cloud Infrastructure

- Supports the heavy processing requirements and data storage with scalable cloud resources, ensuring efficient real-time analytics and integration with digital advertising systems.

How to Carry Out the Project

1. Requirement Analysis and Planning

- Define functional (engagement measurement, anomaly detection) and non-functional (privacy, latency, scalability) requirements.
- Identify integration points with camera and digital signage systems.

2. Data Collection and Preprocessing

Collect camera video footage from retail partners or publicly available retail datasets. Prepare and annotate data as needed for training and validation of pose and anomaly detection models.

3. System Design and Development

- **Pose Estimation and Tracking Modules:** Implement and optimize pose detection using models and incorporate multi-object tracking algorithms to enable identity persistence across frames.
- **Engagement Detection:** Engineer a scoring system that quantifies shopper engagement based on pose-derived features such as body orientation toward ads, dwell time, and stillness. Calibrate and validate scoring with annotated datasets.
- **Implementation of Anomaly Detection:** Develop anomaly detection models leveraging pose sequences to flag suspicious behaviors without using facial or biometric data. Employ unsupervised or semi-supervised machine learning approaches aligned with datasets like PoseLift.
- **Ad Recommendation Integration:** Establish data pipelines for real-time transmission of engagement scores to an adaptive advertising system. Enable dynamic content updates based on live shopper engagement metrics.
- **Security Dashboard:** Show live engagement metrics and anomaly events. Allow operators to respond instantly to security alerts.

4. Testing and Iteration

- Conduct controlled environment testing with varying crowd densities.
- Validate engagement measurement accuracy against ground truth observations.
- Evaluate anomaly detection recall and precision.

5. Deployment and Scalability

- Deploy on a cloud platform for low-latency processing.

Tasks and Sub-Tasks Identified to Achieve Objectives

1. Data Collection and Pose-Based Behavior Assessment

- Task 1.1: Collect anonymized video footage from screen camera for pose estimation and tracking.
- Task 1.2: Preprocess video data by normalizing frame rates, filtering noise, and anonymizing any identifiable information.
- Task 1.3: Annotate datasets with skeletal keypoints and label engagement levels and behavioral anomalies (e.g., suspicious behavior).

2. Pose Estimation and Multi-Person Tracking Module Development

- Task 2.1: Design and implement pose estimation model for real-time multi-person skeletal keypoint detection.
- Task 2.2: Integrate multi-object tracking algorithms to maintain persistent identities for temporal behavior analysis.
- Task 2.3: Optimize models for latency and accuracy in crowded retail environments.

3. Engagement Scoring and Anomaly Detection System

- Task 3.1: Engineer the Engagement Index by fusing pose-orientation, dwell time, and stillness features.
- Task 3.2: Develop machine learning-based anomaly detection models for identifying suspicious or unusual shopper behavior.
- Task 3.3: Validate and refine scoring and anomaly models with annotated datasets.

4. Adaptive Advertisement Engine Integration

- Task 4.1: Build interfaces for secure, real-time transmission of engagement scores to the advertising engine.
- Task 4.2: Implement dynamic ad selection logic based on live engagement metrics to optimize content relevance.
- Task 4.3: Test integration in simulated retail scenarios and refine for responsiveness.

5.Alert System

- Task 5.1: Implement alert generation logic and delivery mechanisms.
- Task 5.2: Conduct usability testing and incorporate user feedback.

6. System Integration, Deployment, and Testing

- Task 6.1: Integrate all modules within a scalable cloud or on-premise infrastructure.
- Task 6.2: Deploy the complete system in a test retail environment.
- Task 6.3: Perform comprehensive system testing covering real-time performance, accuracy, privacy compliance, and user acceptance.
- Task 6.4: Collect feedback from marketing personnel and security teams for iterative enhancements.

Data Requirements and Collection Methods

Data Needed

- Anonymized Pose Data: Labeled datasets containing video frames or sequences annotated with skeletal keypoints (head, shoulders, torso, etc.) of multiple people in retail-like environments for training pose estimation and tracking models.
- Behavioral Anomaly Data: Video clips or pose sequence datasets annotated with typical and atypical behaviors such as loitering, suspicious movements, or potential shoplifting incidents for anomaly detection training.

Data Collection Methods

- Use of Public Datasets: Initially employ publicly available pose estimation datasets and retail behavior video datasets for baseline model training.
- Video Data Collection: Collaborate with retail partners to access anonymized video footage from Screen camera, ensuring privacy compliance by omitting any facial or biometric information.

Use Case Diagram

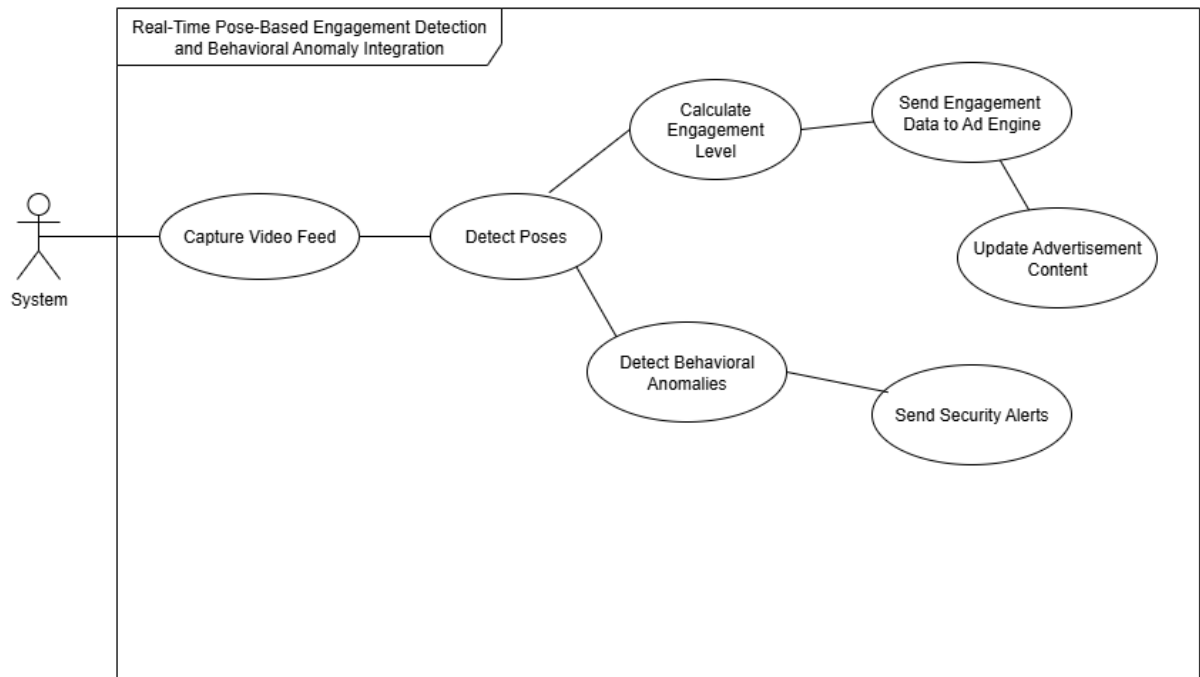


Figure 4 : Use Case Diagram for proposed component

Anticipated Results and Applications

- A real-time, privacy-compliant retail analytics system providing continuous shopper engagement scoring and behavioral anomaly detection.
- Dynamic advertisement content adaptation via integration with advertising engines, enhancing marketing effectiveness.
- Proactive security monitoring through real-time alerts on suspicious shopper behaviors, aiding loss prevention.
- User-friendly analytics dashboard supporting data-driven decisions by marketers and security personnel.
- Overall enhancement in customer experience, operational efficiency, and security in crowded retail environments.

Gantt Chart

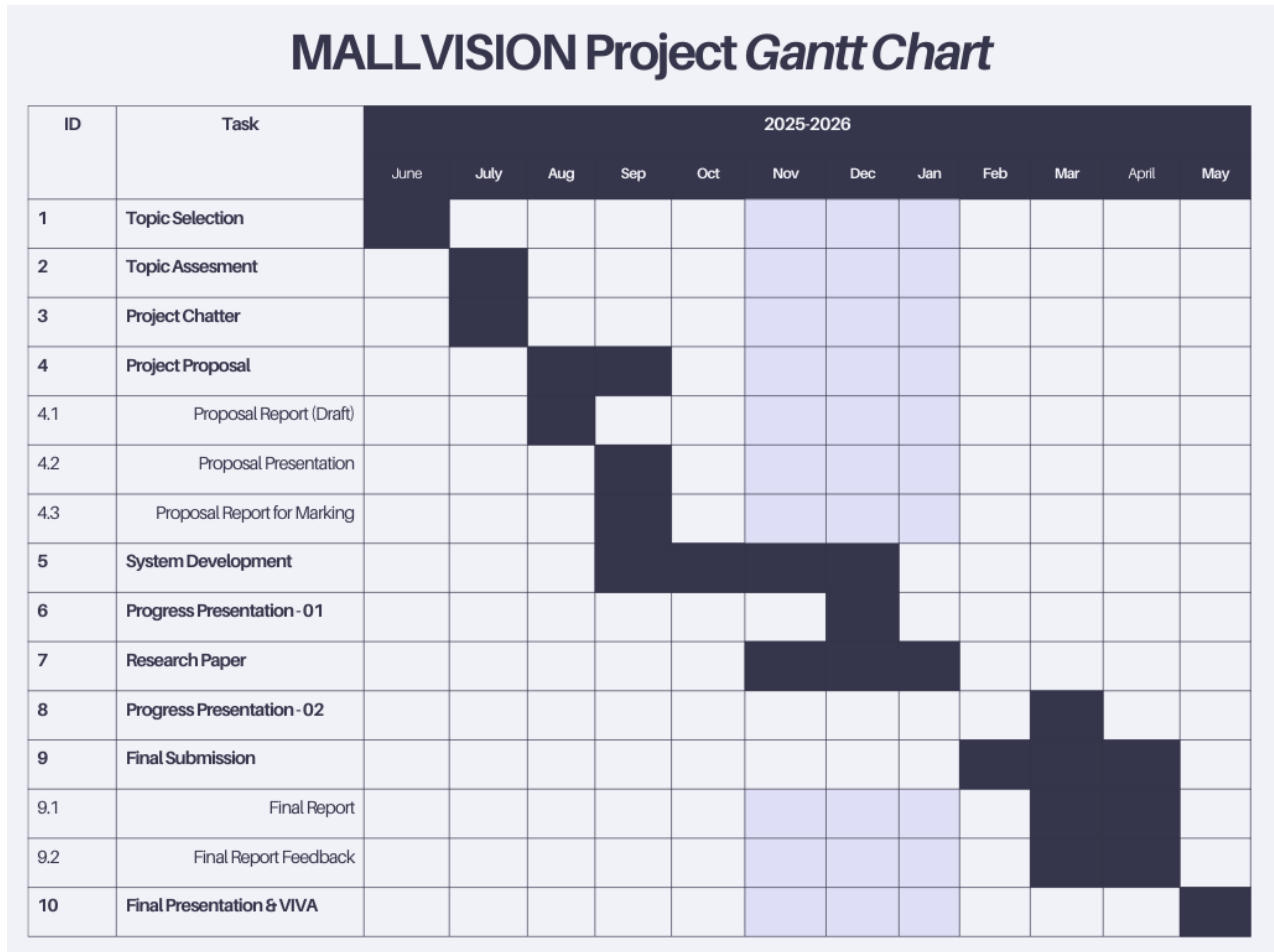


Figure 5 : Gantt Chart

Work Breakdown Structure

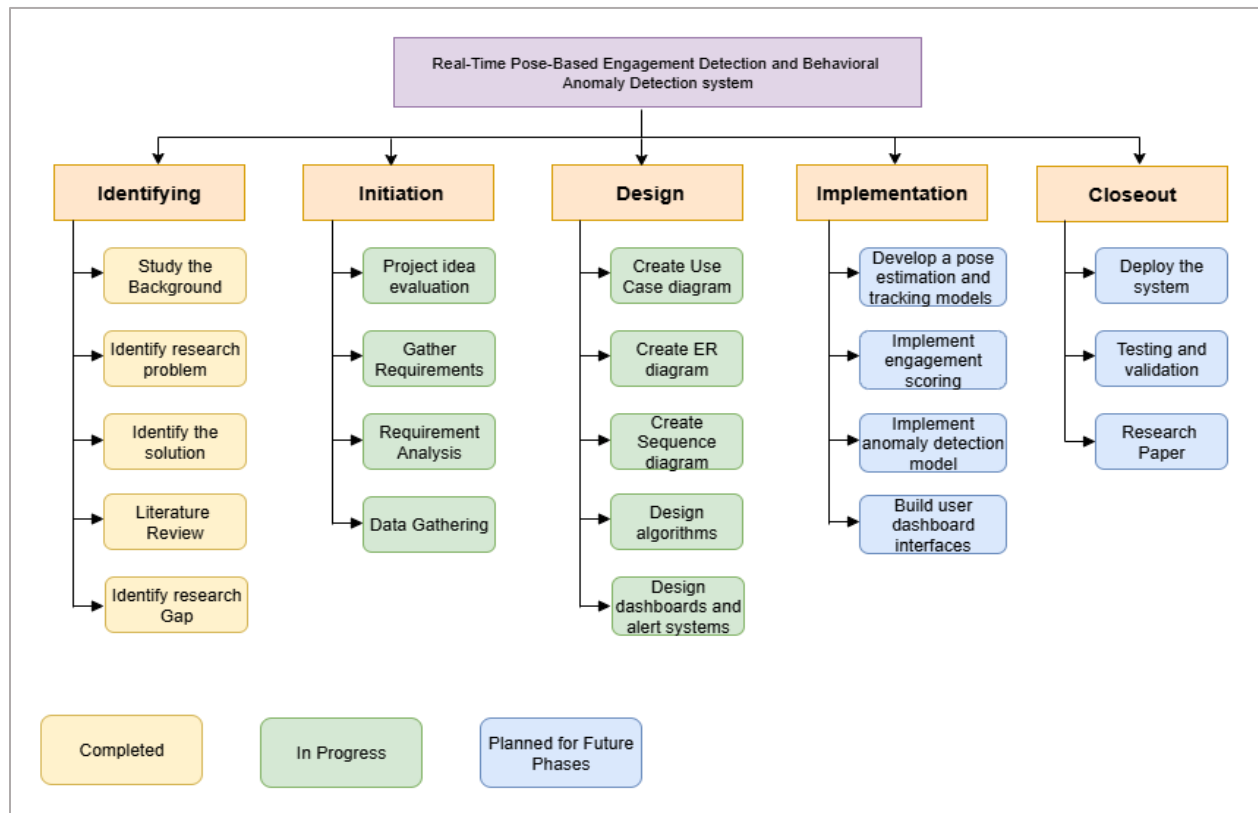


Figure 6 : Work breakdown chart

Commercialization Aspects of the Product

The commercialization potential of this retail analytics product is strong, driven by the rising adoption of AI-driven solutions in the retail sector and growing retailer demand to optimize customer engagement and security in physical stores.

Target Audience (Primary Demographic)

- Retail businesses and shopping mall operators looking to enhance customer experience and security.
- Marketing managers focused on personalized advertising and measuring shopper engagement.
- Security personnel needing real-time behavioral anomaly alerts to prevent loss and ensure safety.

- System administrators managing retail analytics platforms and data privacy.
- Tech-forward retailers adopting AI and cloud solutions for data-driven operations.

Market Space

- **Retail Analytics Market Size:**
The global retail analytics market is projected to be around USD 10-15 billion in 2025, with a strong compound annual growth rate (CAGR) between 17% and 20%. It is expected to reach USD 40-68 billion by 2034-2035 driven by retail digital transformation and AI analytics adoption.
- **Growth Drivers:**
Rising demand for real-time data analysis, recommendation engines, customer behavior insights, and loss prevention solutions.
- **Trends:**
Increasing use of multi-object tracking, behavior analysis, adaptive advertising, and cloud-based solutions bolster market growth. Retailers seek advanced analytics to optimize inventory, personalize marketing, and enhance security.
- **Competitive Landscape:**
While many retail analytics providers exist, your product's USP in real-time pose-based engagement detection combined with behavioral anomaly identification and privacy-preserving AI offers a unique niche for differentiation.

Unique Selling Proposition (USP)

1. Privacy-First AI Analytics.
2. Real-Time Content Adaptation
3. Dual-Purpose Functionality
4. Plug-and-Play Integration

Description of Personnel and Facilities

Facilitators

- Dr. Bagya Nathali Silva – Sri Lanka Institute of Information Technology (SLIIT)
- Ms. Shashini Kumarasinghe – Sri Lanka Institute of Information Technology (SLIIT)

Member Contribution

The contributions to this project involve a comprehensive approach to developing a privacy preserving, Dynamic advertising system that dynamically adapts content in shopping malls based on real time audience demographics, emotional state and behavioral analytics. Fonseka W S M focuses on audience demographic detection using computer vision models to estimate age groups in real time while ensuring that no identifiable facial data is stored, thus maintaining privacy compliance. Samaraweera W D U I works in emotion detection using computer vision models to identify emotions such as happy, sad, angry and neutral in real time and detect suspicious or ignoring behaviors for security alerts also handling occlusions to maintain accurate detection. Abeykoon S N works on pose-based engagement detection and behavioral anomaly analysis, calculating the Engagement Index from body orientation and dwell time, and also identifying unusual behaviors for both marketing insights and security alerts. Rajana H T R develops dynamic advertisement recommendation engine, integrating demographic, emotional and behavioral data into a single pipeline that updates mall display content in real time to increase engagement.

Project Requirements

Functional Requirements

- **Real-Time Video Capture**

This component continuously receives video feeds from cameras installed throughout the shopping mall. The video feed is streamed in real time to a central processing unit or server capable of handling high-throughput video input. This live video capture forms the input foundation for downstream pose estimation and behavior analysis modules.

- **Multi-Person Pose Estimation**

Using the real-time video frames from camera, this module applies advanced deep learning-based pose estimation algorithms to detect and extract skeletal keypoints of all visible shoppers in the scene. The system extracts coordinate of key body joints like head, neck, shoulders, elbows and torso —strictly avoiding facial recognition or demographic features to maintain privacy. The model is designed to identify and separate multiple individuals simultaneously within crowded retail environments.

- **Real-Time Pose Detection**

The system must use advanced computer vision algorithms (e.g., pose estimation) to detect skeletal key points of multiple individuals in the camera feed. It should determine body orientation and dwell time for each detected individual. These metrics will be combined into an Engagement Index to quantify audience attention toward advertisements.

- **Engagement Index Calculation**

The system should measure engagement levels based on body orientation and dwell time near the display screens.

E-Index should be calculated per person and averaged for the current audience to influence advertisement selection in real time.

- **Behavioral Anomaly Detection**

The system must detect unusual movement patterns, such as loitering, sudden dispersal, or erratic walking, to provide additional security alerts.

- Detected anomalies should be logged and made available to security personnel through the dashboard.
- **Dynamic Ad Recommendation Integration**
The system should send real-time audience metrics to the advertisement recommendation engine. Ads must be updated in real-time according to audience engagement level and behavior.
- **Privacy-Preserving Processing**
All detection must be performed without storing biometric identifiers such as facial images. The system should only store anonymized skeletal and movement data.

Non-Functional Requirements

• Data Privacy

- Ensure compliance with GDPR and other relevant privacy laws.
- Use on-device anonymization and encryption for all engagement/anomaly data.

• Accuracy

- Engagement detection accuracy $\geq 90\%$ in optimal lighting.
- Anomaly detection false-positive rate $\leq 5\%$.

• Performance

- Maximum latency from video capture to output: ≤ 1 second.

• Reliability

- Maintain 99% uptime during operational hours.

User Requirements

• Marketing Team

- View real-time engagement metrics and reports.
- Understand which ads are performing best for different audience profiles.

Security Team

- Receive instant alerts for unusual behaviors.
- Access anonymized footage for verification.

• System Administrator

- Configure camera feeds, detection thresholds, and alert settings.

System Requirements

Hardware

- High-performance servers or GPUs capable of real-time video processing and deep learning inference.
- Camera systems providing video feeds.

Software

- Deep learning frameworks (e.g., TensorFlow or PyTorch) for Pose estimation anomaly detection
- Multi-object tracking libraries
- Web-based analytics dashboard.
- Secure API or messaging protocols for data transmission between module

Network

- Robust and secure network infrastructure to support real-time data transfer from cameras to servers and between system components.

Data

- Access to anonymized retail camera video datasets for model training and validation

Tentative Test Cases

1. Engagement Index Calculation

Test Case 1.1 – Basic Pose Detection

- **Objective:** Verify system detects skeletal keypoints in real-time.
- **Input:** Video feed with multiple individuals.
- **Expected Output:** All persons detected with accurate body orientation.

Test Case 1.2 – Dwell Time Tracking

- **Objective:** Ensure dwell time is correctly measured.
- **Input:** Person stands facing ad for 10 seconds.
- **Expected Output:** Dwell time recorded as ~10 seconds.

Test Case 1.3 – Engagement Score Accuracy

- **Objective:** Confirm E-Index is computed correctly.
- **Input:** Multiple people with varying dwell times.
- **Expected Output:** Higher scores for longer, direct attention.

2. Behavioral Anomaly Detection

Test Case 2.1 – Loitering Detection

- **Objective:** Verify loitering is detected after predefined threshold.
- **Input:** Person remains in a zone > 60 seconds.
- **Expected Output:** Security alert triggered.

Test Case 2.2 – Erratic Movement Detection

- **Objective:** Detect sudden, irregular movement patterns.
- **Input:** Simulated running in random directions.
- **Expected Output:** System flag as anomaly

Budget and Budget Justification

The proposed budget covers essential expenses required to develop, deploy, and validate the smart real time age and emotion-based advertising system in shopping malls using the custom-built models. Following is the budget estimation for our whole research system.

Table 2 : Budget Plan

Requirement	Cost (Rs.)
Hardware Cost	20,000.00
Travelling Cost	5,000.00
Software and Subscription Costs	6,000.00
Total	31,000.00

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